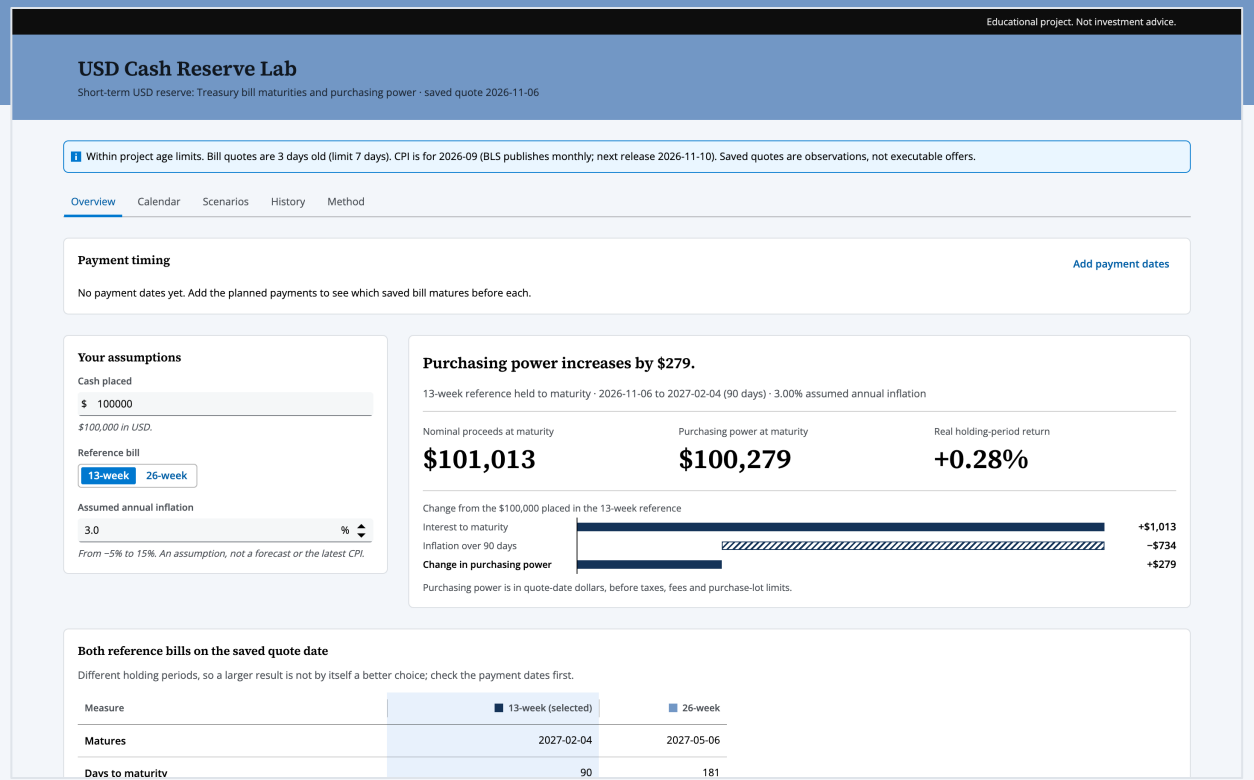
Backup slides

# USD Cash Reserve Lab

When is the cash there for each payment,
and what does inflation do to it meanwhile?




**Group 9** · BMS5312 Fintech Management
Live dashboard demo, 5 minutes

Educational project. Not investment advice.

Speaker 1 · The user and the question

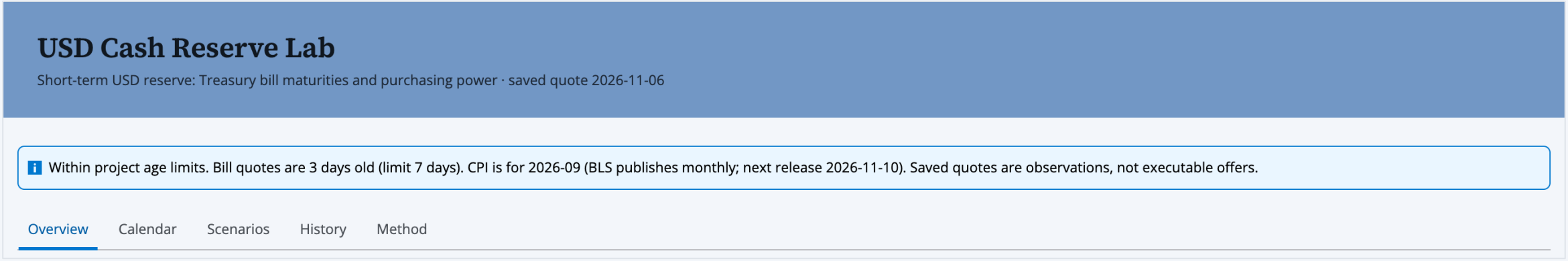
# Who is the user, and what decision does the page support?

## The user

- A family-office client and their advisor
- A US-dollar reserve for payments in the next few months: tax, tuition, a household budget
- **Today:** spreadsheets and separate data websites

## What the page does

- Shows which saved Treasury bill matures in time for each payment, and what an assumed inflation rate does to the cash meanwhile
- Liquidity first, so no bill is ranked or recommended
- Official data, dated on the page (U.S. Treasury daily bill rates, BLS CPI-U)



Educational project. Not investment advice.

Speaker 2 · Live dashboard

# When is the cash there for each payment?

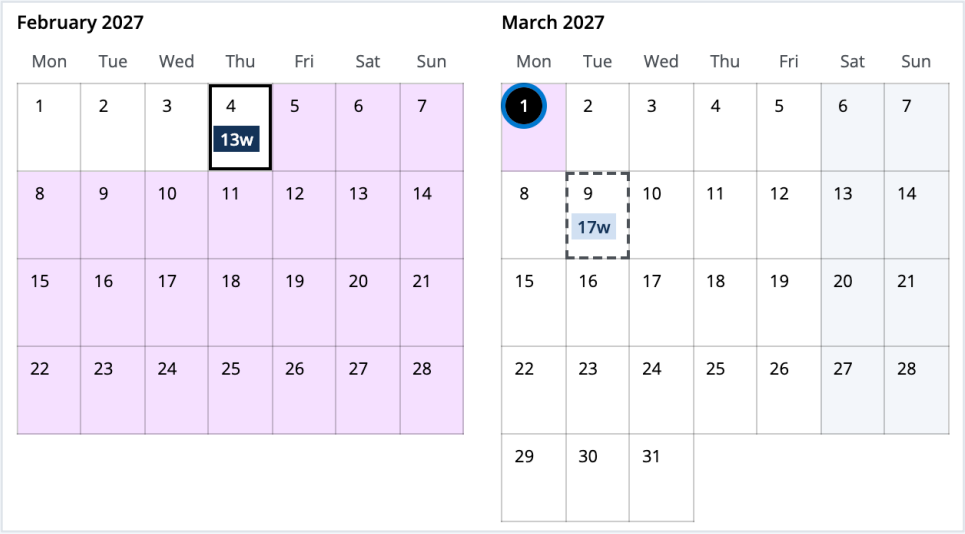
## Tuition, Monday 1 March 2027

Fixed USD payment

13w **Matures on or before:** 13-week bill, 2027-02-04

**Cash waits 25 days**, no return assumed

17w **Next maturity:** 17-week bill, 2027-03-09, 8 days late
So it would have to be sold before maturity at an unknown price




## Calendar fit by payment

| Payment | Date | Matures on or before | Cash waits | Next maturity after |
|---|---|---|---|---|
| Property tax instalment | 2027-01-15 | 8-week · 2027-01-05 | 10 days | 13-week · 2027-02-04, 20 days after |
| Tuition | 2027-03-01 | 13-week · 2027-02-04 | 25 days | 17-week · 2027-03-09, 8 days after |
| Spring household budget | 2027-04-15 | 17-week · 2027-03-09 | 37 days | 26-week · 2027-05-06, 21 days after |

Speaker 2 · Live dashboard

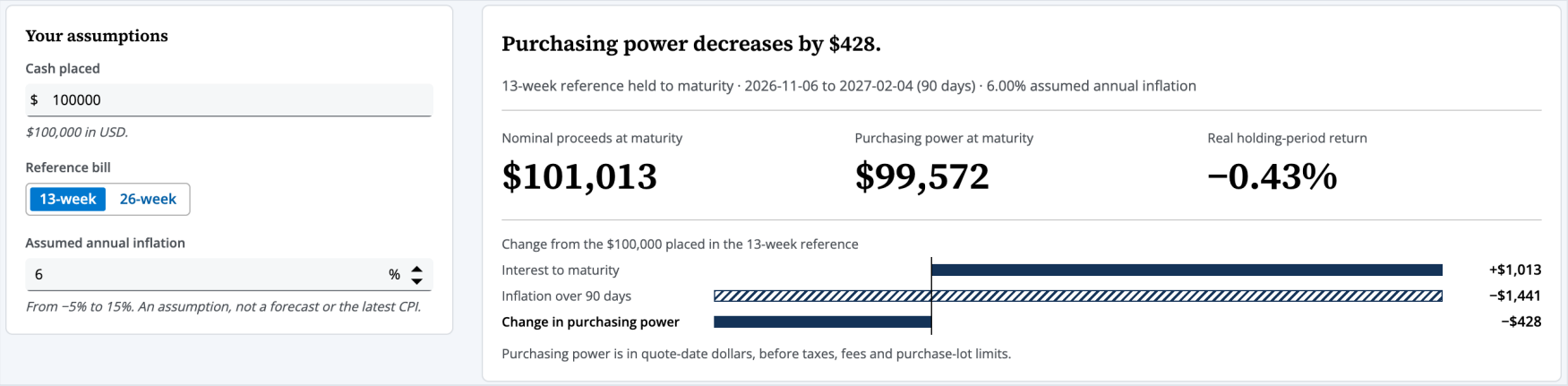
# What does inflation do to the cash?

**$100,000 in the 13-week bill** · 2026-11-06 to 2027-02-04 (90 days) · bank discount quote 4.01%

| | Inflation 3% | Inflation 6% |
|---|---|---|
| Nominal proceeds at maturity | $101,013 | $101,013 |
| Purchasing power at maturity | $100,279 | $99,572 |
| Real holding-period return | +0.28% | −0.43% |
| **Change in purchasing power** | **+$279** | **−$428** |

> Inflation changes what the money buys, not the dollars that arrive.

**Break-even inflation:**
13-week 4.17%, 26-week 4.31%.
A threshold, not a forecast.




Live page, inflation set to 6%

Speaker 3 · Data and method

# How do we know the numbers are right?

Price per $100

$$P = 100\,(1 - d \cdot n/360)$$

Nominal proceeds

$$N = B \cdot 100/P$$

Inflation factor

$$F = (1 + a)^{n/365}$$

Purchasing power

$$R = N/F$$

## Independent Excel check

By a member who did not build the page · quote 2026-11-06 · inputs copied from the official websites

| Check | Excel | Page | Result |
|---|---|---|---|
| Price per $100 | 98.9975 | — | *input to the next rows* |
| Nominal proceeds | 101,012.65 | $101,013 | **PASS** |
| Purchasing power at 3% | 100,279.10 | $100,279 | **PASS** |
| Real holding-period return | 0.279% | +0.28% | **PASS** |
| Break-even, 13-week | 4.1708% | 4.17% | **PASS** |
| Break-even, 26-week | 4.3114% | 4.31% | **PASS** |
| CPI year on year, Sep 2026 | 3.7377% | 3.74% | **PASS** |

**6 / 6**
checks passed

## Limitations

- Saved quotes are observations, not executable prices.
- Settlement, taxes and fees are not modelled.
- Inflation is an assumption, not a forecast.

Speaker 4 · AI maintenance

# How does an AI assistant update the data, and who decides?

**AI assistant**

1. Fresh AI chat reads SKILL.md
2. Downloads into candidate/
3. Rebuilds and pre-reviews
4. Before/after table with source links

**Person**

5. Person compares and accepts
6. Release; old version kept; checks rerun

## Normal update, 2026-11-09

| | Before | After |
|---|---|---|
| Treasury quote date | 2026-10-26 | **2026-11-06** |
| 13-week bank discount | 4.05% | **4.01%** |
| 26-week bank discount | 4.21% | **4.12%** |
| CPI month | 2026-09 | **2026-09** (no new release) |

## Failure test

Rates set to basis points
→ **STOP: Incompatible input unit**
Nothing written; data.js unchanged.

The AI does the checks; the source comparison and the decision stay with a person.

Speaker 5 · Handover

# Can someone else run it after the course?

## Handover

- The ZIP alone works
- Double-click index.html
- One command rebuilds data.js byte for byte
- One command downloads and checks an update
- Nothing is released until a person accepts
- The previous version is one command away

## Risks

- No scheduler, so quotes go stale after 7 days and the page says “Review required”
- BLS release dates must be added each December
- October 2025 CPI was never published

**TO REPLACE AFTER THE USER TEST**

User test by a member who did not build the page:
what they found, and the one change we made.

## Next improvement

Add payment amounts to the calendar,
so it shows whether the cash covers each payment,
not only the date.